

Chapter 7

Mobile Application Packaging and
Publication

Contents

Checklist to plan your app launch

Prepare for release

App Bundles

Android Go

Google Play Instant

Support Chrome OS

Monetize your app

Checklist

1. Developer Program Policies
2. Developer Account
3. Localization
4. Device Compatibility
5. Quality Test : Alpha & Beta
6. Store Listing

Check List

1. Developer Program Policies

Mobile app stores, such as the Google Play Store, Apple App Store and Huawei App Gallery, have specific guidelines and policies that developers must adhere to

These policies are designed to maintain a safe, secure, and positive user experience.



Check List - Developer Program Policies



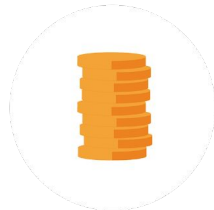
Restricted
Content



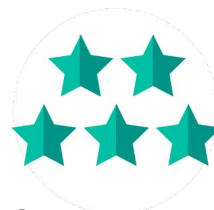
Intellectual
Property



Privacy and
Security



Monetization
and Ads



Store Listing
and
Promotion

Check List - Developer Program Policies

Mobile app stores have strict rules about the type of content allowed, including explicit content, malware, and harmful apps.

Explicit Content: pornography, sexually suggestive content, and nudity



Check List - Developer Program Policies

App reviewers check for potential IP infringement, such as copyright infringement, trademark infringement, and patent infringement.



Intellectual
Property

Check List - Developer Program Policies

Developers must have clear privacy policies that explain how user data is collected, used, and shared

App stores may impose security requirements to protect user data from unauthorized access and breaches



Privacy and
Security

Check List - Developer Program Policies

App stores provide guidelines govern in-app purchases and subscriptions

Ad monetization and in-app purchases must adhere to guidelines, including fair pricing and transparent advertising.



Monetization
and Ads

Check List - Developer Program Policies

App stores control the listing and promotion (marketing materials) of apps through strict review processes, guidelines, and algorithms

Developers must adhere to these rules to ensure their apps are visible, accessible, and successful. E.g. setting of user's age groups, regions, etc

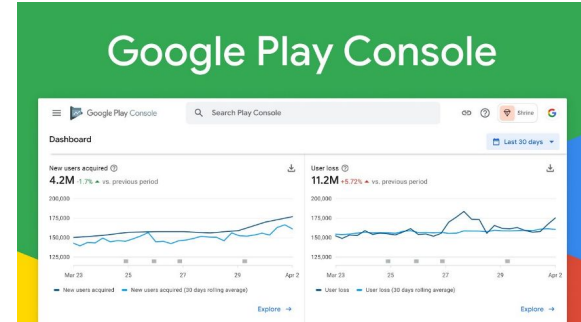


Store Listing
and
Promotion

Check List

Developer Program Policies

Developer Account



Developer Account

It is a **publishing account** issued by **Platform Provider** to **developer**

It enables developer to **post, display, offer for sale, and distribute apps** through the Platform

Access developer tools, resources, and support provided by the platform.

Developer Account



USD 25
(one time)



USD 99
(per year)



FOC

KaiOS

FOC



FOC

 All apps

 Inbox 16

 Policy status

 Users and permissions

 Order management

▶  Download reports

 Account details

 Developer page

 Associated developer accounts

 Activity log

▶  Setup

All apps

View all of the apps and games that you have access to in your developer account

Create app

Pinned apps ?

Pin apps here to access them quickly and view key metrics

4 apps

Filter by

All

App	Installed audience	App status	Update status	Last updated		
 GBA Market Tracker com.mt.gba.gbamarkettracker	33	Production		6 Oct 2019		
 Multilingual Storybook mmsr.mystorybook	1	Removed by you	 In review	9 Sep 2021		
 TAR UC Bus my.edu.tarc.testing2	0	Removed by you		12 Oct 2017		
 TAR UC Bus Module (Alpha) com.wy.wenyang.tarucbustracking...	0	Removed by you		3 Nov 2016		

☰ Welcome

Additional Resources

📖 Documentation

⬇ Downloads

💬 Forums

💡 Feedback Assistant

❓ Account Help

📞 Contact Us

Getting Started

Download Xcode, learn how to build an app, and install it directly on your Apple device.



Download Tools

Get started with Xcode, Apple's integrated development environment for creating apps.



Build Your First App

Use Xcode to write a simple "Hello World" app to get familiar with tools, SDKs, and the Swift language.

Join the Apple Developer Program

Membership in the Apple Developer Program includes everything you need to develop, distribute, and manage your apps on the App Store. You'll also gain access to beta software, advanced app capabilities, beta testing tools, and app analytics.

Ecosystem services



App services



Content services

HMS API services



My APIs



API Library



Credentials



Privacy contact info

Developer center



My reports



Customer service



My messages



My accounts



Settings

Distribution and promotion

AppGallery Connect

Your one-stop open platform covers the entire app lifecycle: innovation, development, distribution, operation, and analysis.

Monetize Service



Publisher Service

Development



Push Kit



HUAWEI IAP



HUAWEI ID



HUAWEI HIAI



Health Kit



Wear Engine



HarmonyOS Development Agreement

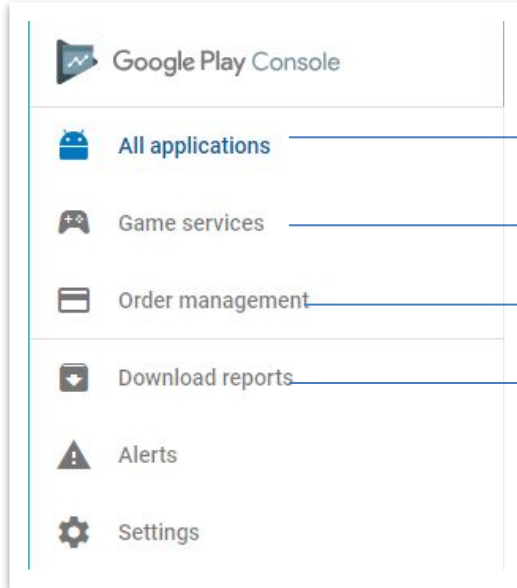


3D Object Recognition

New

Developer Account

Google Play Console - Features



View your apps, active installs, rating, last update and status

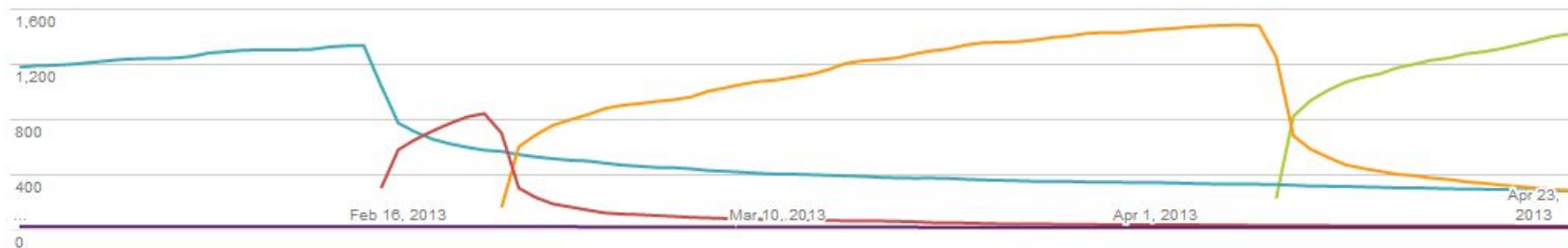
Add social gaming features to your games

Earn money with paid apps or in-app products

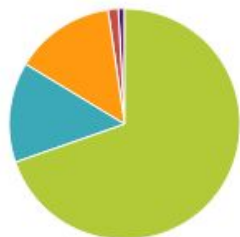
Know your app's performance

Developer Account

ACTIVE DEVICE INSTALLS BY APP VERSION



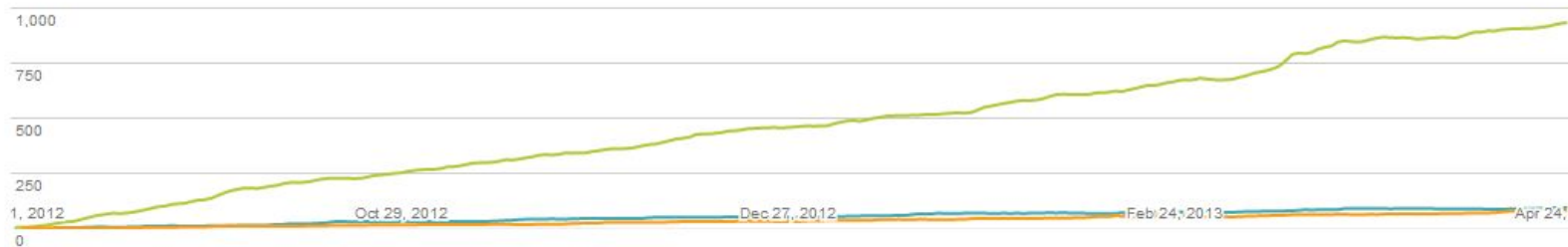
ACTIVE DEVICE INSTALLS ON APR 25, 2013



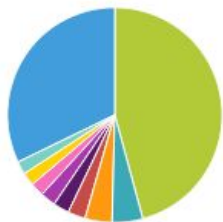
YOUR APP		
☒ 36	1,421	69.66%
☒ 21	286	14.02%
☒ 34	286	14.02%
☒ 31	29	1.42%
☒ 18	18	0.88%

Developer Account

ACTIVE DEVICE INSTALLS BY COUNTRY



ACTIVE DEVICE INSTALLS ON APR 25, 2013



	YOUR APP	
☒ United States	934	45.78%
☒ Canada	94	4.61%
☒ United Kingdom	83	4.07%
☐ South Korea	54	2.65%
☐ Germany	49	2.40%
☐ France	47	2.30%
☐ Australia	42	2.06%
☐ Indonesia	40	1.96%
☐ India	40	1.96%
☐ Others	657	32.21%

ALL APPS IN TOOLS
23.86%
1.33%
2.94%
9.64%
3.88%
2.86%
1.12%
1.10%
2.71%

TOP 10 COUNTRIES FOR TOOLS	
United States	23.86%
Japan	11.25%
South Korea	9.64%
Germany	3.88%
Russia	3.06%
Spain	2.97%
United Kingdom	2.94%
France	2.86%
India	2.71%
Brazil	2.52%

Check List

Developer Program Policies

Developer Account

Localization



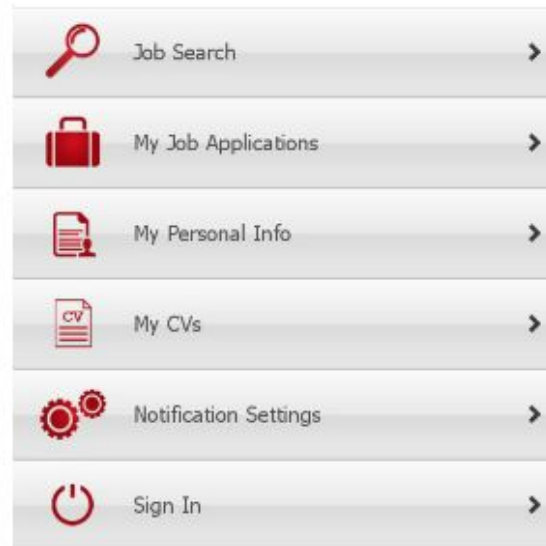
Localization



The adaptation of an app to meet the needs of a particular **language**, **culture** or desired population's "look-and-feel."

A successfully localized app is one that appears to have been developed within the **local culture**.





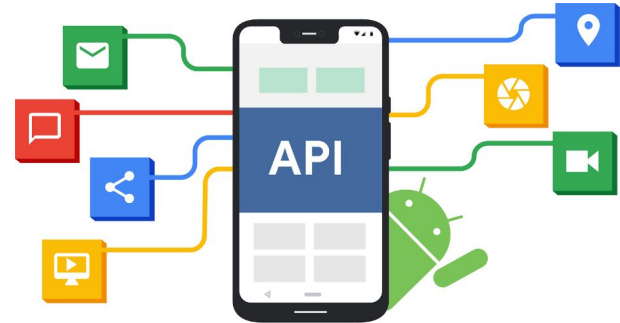
Check List

Developer Program Policies

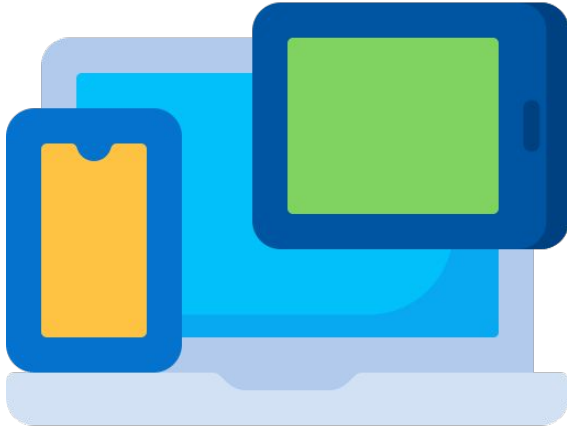
Developer Account

Localization

Device Compatibility



Device Compatibility



It ensures that an app runs efficiently **across mobile devices** of **different configurations**

Compatibility also should ensure app run well on mobile devices of different **hardware**, **software**, and **operating systems**

Check List

Developer Program Policies

Developer Account

Localization

Device Compatibility

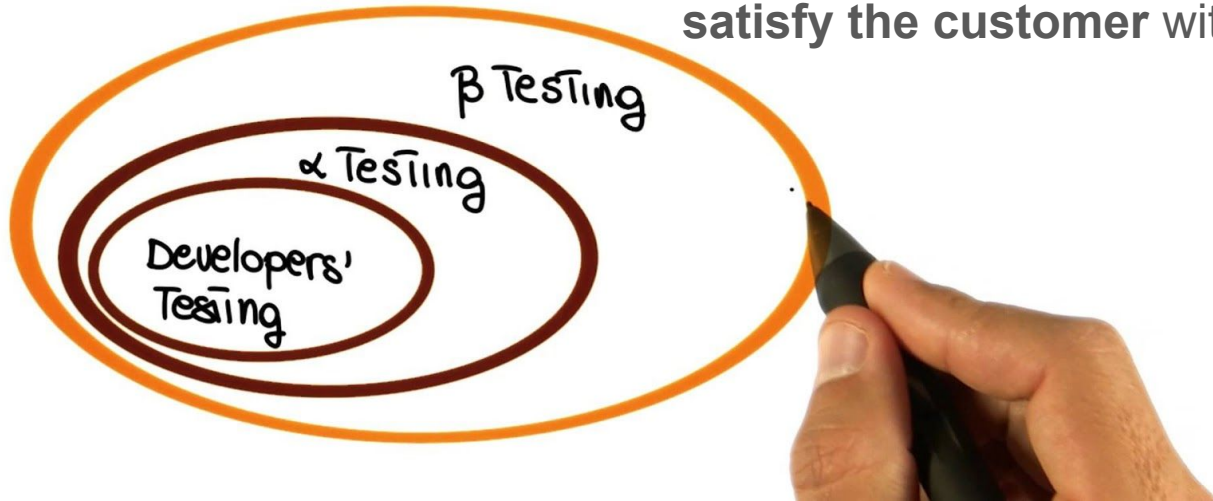
Quality Test : Alpha & Beta



Quality Test

A process to ensure apps **meet specified regulations and standards**.

It's a series of techniques that developers employ to **prevent issues from occurring** and **ensure they satisfy the customer** with their finished product.



Check List

Developer Program Policies

Developer Account

Localization

Device Compatibility

Quality Test; Alpha & Beta

Store Listing



Preparing for Release



To prepare your app so that users can install and run the app on their Android-powered devices

Release-ready .apk file is signed with your own certificate

Preparing for Release

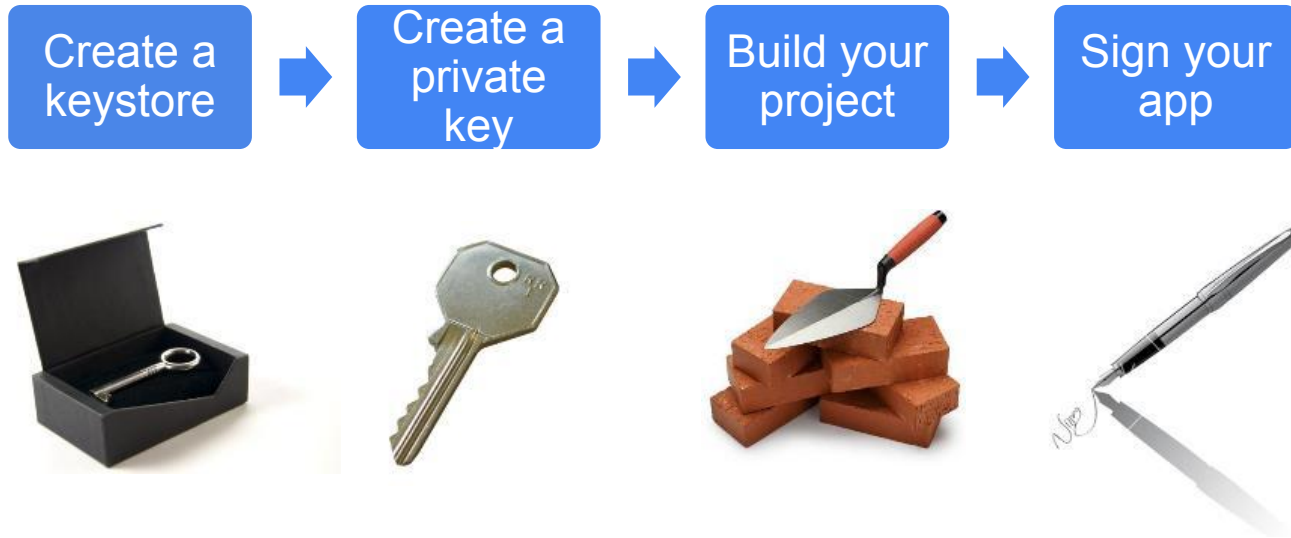


Minimum requirements:

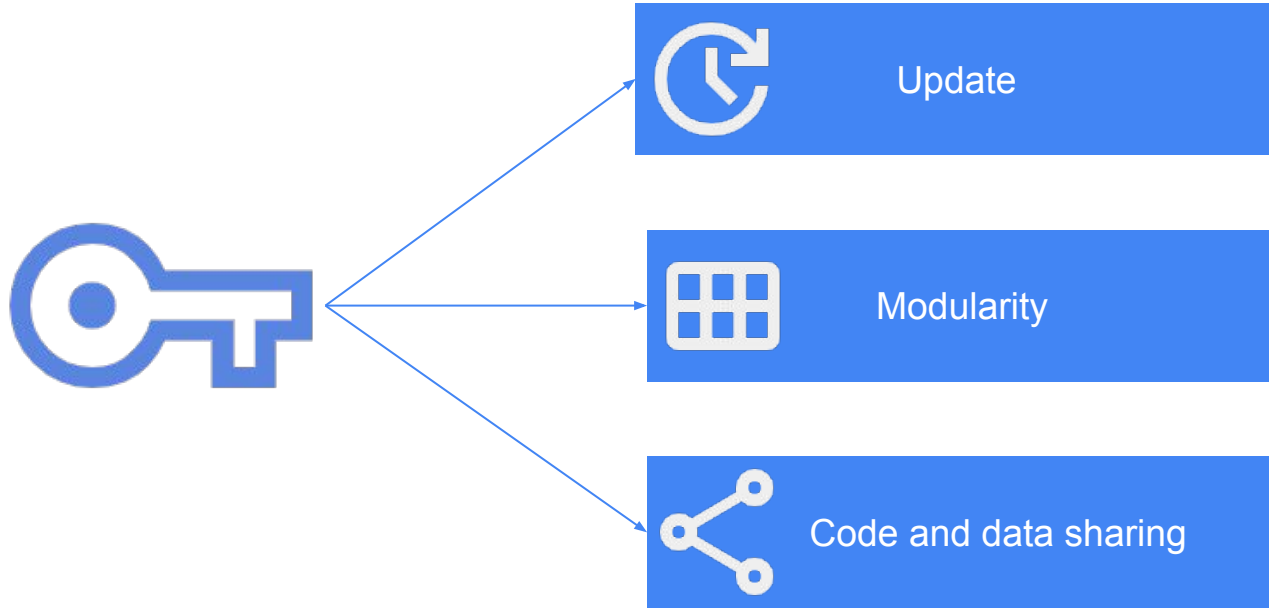
1. Cryptographic keys
2. Application icon
3. End-user License Agreement
4. Promotional and marketing materials

Cryptographic key: digitally signature that is owned by the application's developer

Signing in Release Mode



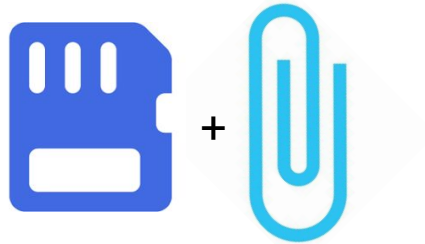
Signing Consideration



Store Listing



Distribution

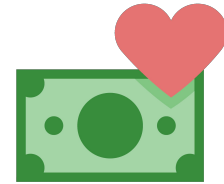


App size
<150 MB

Expansion
up to 2GB

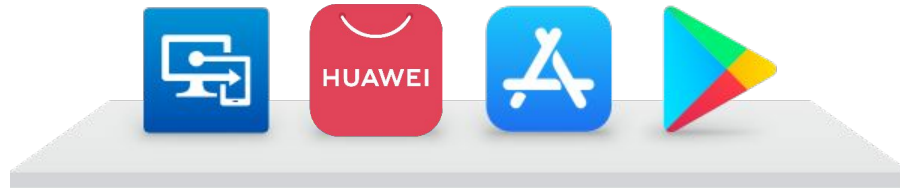


Platforms



Fee or Free

Distribution



Distribution Channels



Marketplace



Email



Website or server

Distribution Methods - Android

1. Android Go
2. App Bundles
3. Google Play Instant
4. Support Chrome OS

Android Go



Android Go is an OS designed for entry-level smartphones with limited resources

An initiative that brings the benefits of Android to a wider range of users

Android Go is optimized for devices with:

- Limited RAM, typically 1GB or less
- Lower processing power
- Less storage space

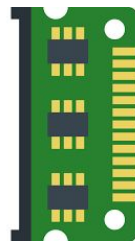
Android Go



Target Oreo
(API 26)



App size less
than 40 MB



RAM usage
below 50 MB
(apps)
150 MB
(games)



Start your
app under
5 seconds

App Bundles

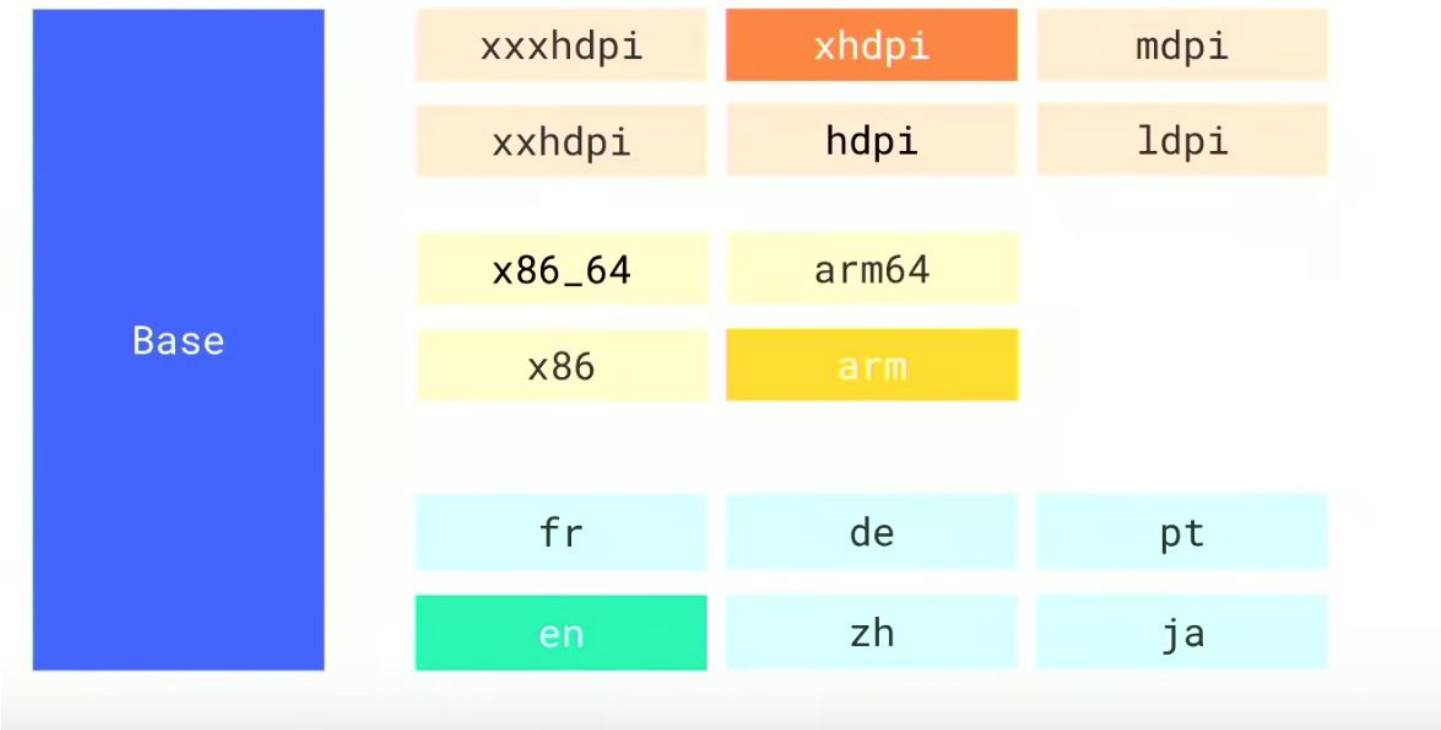


An efficient distribution technology for Android apps, especially on devices with varying screen sizes and capabilities

Instead of building separate APKs for each device configuration (screen size, CPU architecture, etc.), you create a single .aab (Android App Bundle) file.

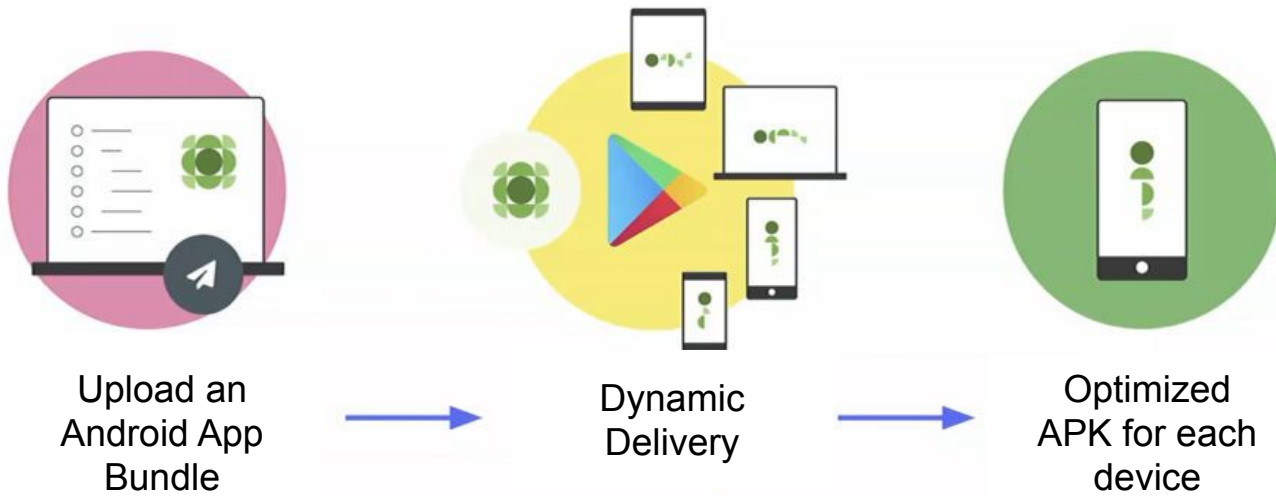
App Bundles

This bundle contains all the compiled code and resources for your app.



App Bundles

Google Play then dynamically delivers only the necessary code and resources for a specific user's device during installation.



App Bundles



Benefits:

1. Small app file: Users download only the parts of your app they need, resulting in significantly smaller download sizes.
2. Fast installation: Faster installation times due to smaller downloads.
3. Good performance: Optimized for specific devices, leading to better performance and potentially reduced battery consumption.
4. Easy to manage: Easier to manage app releases and updates.

Google Play Instant

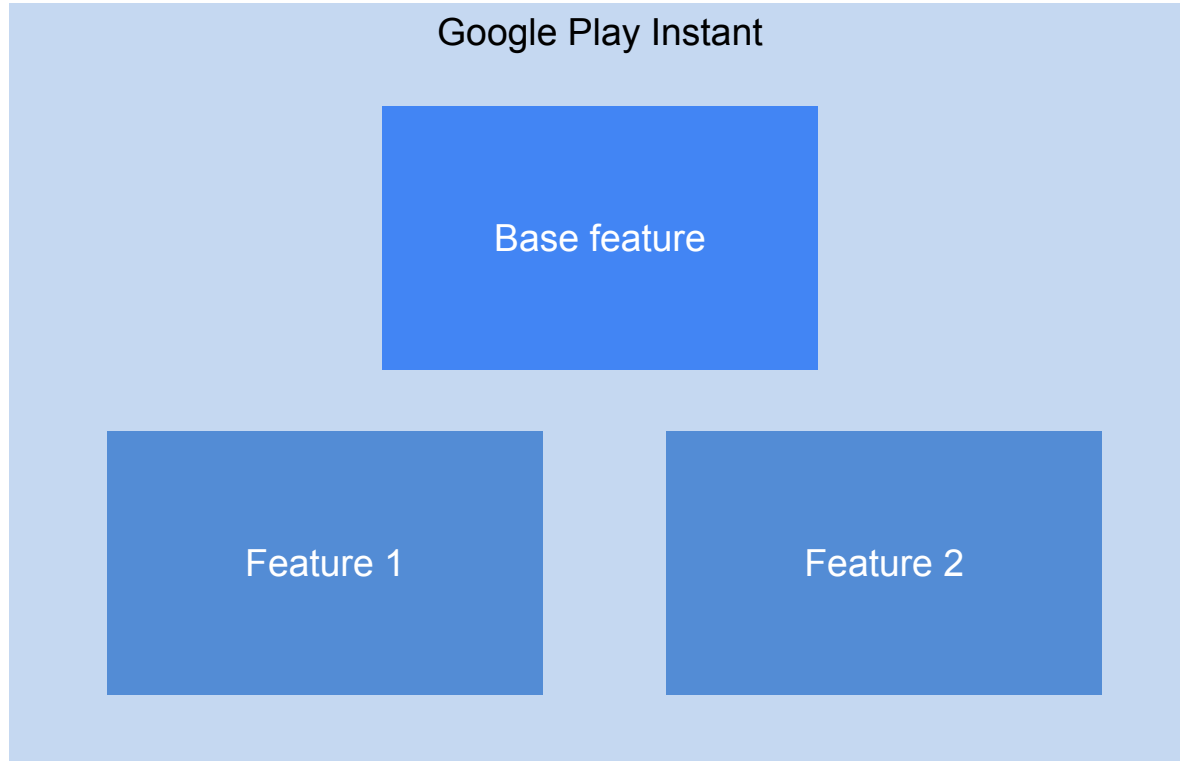


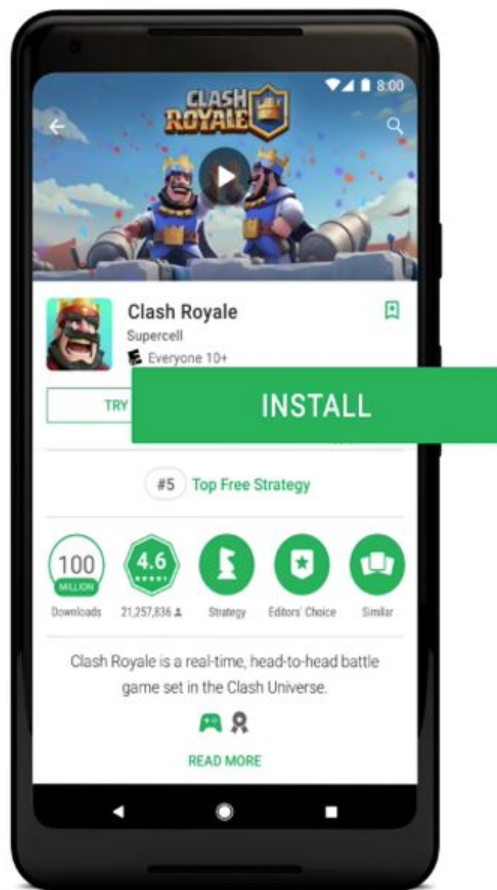
Google Play Instant allows users to try a portion of an app or game before downloading the full version

Users can access instant apps through links in search results, emails, ads, or social media

App has limited functionality, typically focuses on core app functionality or a specific game level

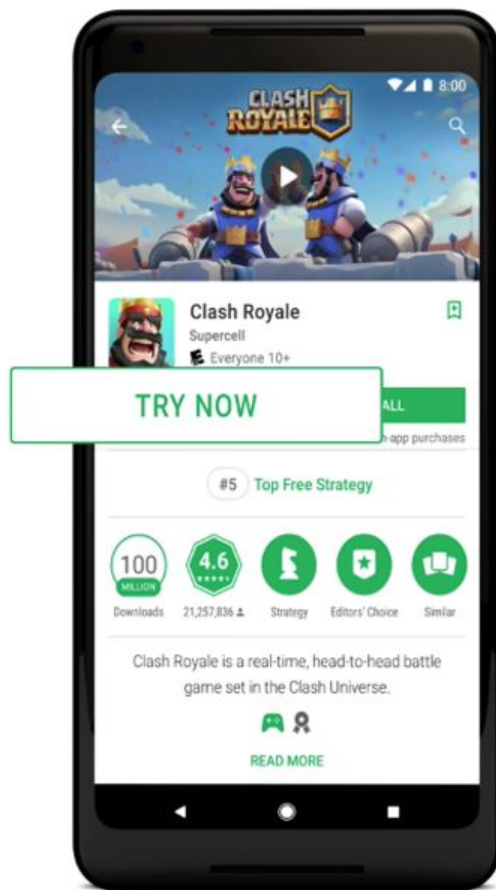
Google Play Instant





Download

3



Instant

Google Play Instant



Benefits

Try Before You Buy: Users can experience the app before committing to a full download.

Reduced Storage Space: No need to install the full app to try it out.

Faster Access: Instant apps load quickly, allowing users to start using them immediately.

Google Play Instant

Limitations



Size Limitations: Instant apps have size restrictions to ensure fast loading times.

Functionality Limitations: Only a subset of the app's functionality may be available in the instant version.

Chrome OS

It is an operating system developed by Google that is designed to work primarily with web applications

It uses the Chrome web browser as its main user interface.



Chrome OS

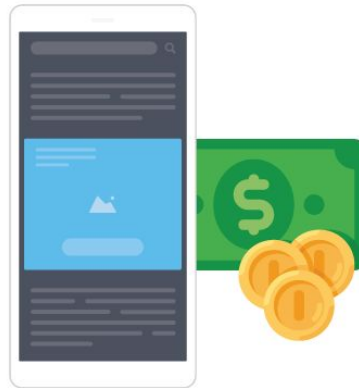
It supports the Google Play Store and Android apps

Preparation:

- Manifest file

```
<uses-feature  
  android:name="android.hardware.touchscreen"  
  android:required="false" />
```

Monetize Your App



Monetize Your App

Monetizing your app is how you generate revenue from it



Developing a mobile app involves various costs:

1. Development costs
2. Tools and technologies
3. Marketing and promotion
4. Ongoing costs - updates, server hosting, customer support, etc
5. Legal and regulatory costs

Monetize Your App



Factors that influence the monetization model:

1. Who are the users? Age, income, tech savviness
2. How valuable is your app? Does it solve a real problem?
3. What are your competitors doing?
4. How much does it cost to make and maintain the app?
5. How will it impact user experience?

Monetize Your App

Who is paying?

Premium Apps

Freemium Apps

Subscription



End-users

Ads

E-Commerce

Rewarded Products

Service

Data Collection



Merchants

Monetize Your App

Premium Apps

Paid

In-app Billing

Extensive Features

Narrow niche in the market



Monetize Your App

Freemium Apps

Free

Uses In-app Billing

Digital goods

- Durable
- Consumable



Monetize Your App

Subscription

Free trial

In-app billing

Subscription fee



Monetize Your App

Ads

Free

AdMob + Google
Mobile Ads SDK

Shows
income-generating
ads



Alternative Monetization Options

E-Commerce

Free

B2C

Technology,
logistics and
payment solutions

Sales commissions
+ Setup fees



Alternative Monetization Options

Rewarded Products

Free

Share content on social media, perform simple tasks (e.g. scan QR code, answer survey and etc)

Get free stuff, earn rewards



Shopback



GigaGigs



Google Opinion
Rewards

Alternative Monetization Options

Service

Free

An extension of
physical/online
services



Alternative Monetization Options

Data Collection

Free

Partnership/Affiliate

Provide customer service, promotion, run a giveaway and etc



Find Out More

Build and release for Android, <https://docs.flutter.dev/deployment/android>

Build and release for iOS, <https://docs.flutter.dev/deployment/ios>

Build and release for Huawei App Gallery,
<https://developer.huawei.com/consumer/en/appgallery/>